

PROBLEM STATEMENT

Fractional Power Composite Rule

Find the derivative with respect to x of the function:

$$y = (2x^{3/2} - 3x^{4/3} - 5)^{5/2}$$

STEP 1 প্রদত্ত ভগ্নাংশীয় ঘাত ফাংশনটি উল্লেখ করি (Given Fractional Power Function)

$$y = (2x^{3/2} - 3x^{4/3} - 5)^{5/2}$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [(2x^{3/2} - 3x^{4/3} - 5)^{5/2}]$$

STEP 3 শক্তি বিধি ও চেইন রুল প্রয়োগ করি ($\frac{d}{dx}[u^n] = nu^{n-1} \frac{du}{dx}$)

$$\begin{aligned}\frac{dy}{dx} &= \frac{5}{2}(2x^{3/2} - 3x^{4/3} - 5)^{\frac{5}{2}-1} \cdot \frac{d}{dx}(2x^{3/2} - 3x^{4/3} - 5) \\ \frac{dy}{dx} &= \frac{5}{2}(2x^{3/2} - 3x^{4/3} - 5)^{3/2} \cdot \frac{d}{dx}(2x^{3/2} - 3x^{4/3} - 5)\end{aligned}$$

STEP 4 বকুনীভূত প্রতিটি ভগ্নাংশীয় ঘাতের অন্তরক নির্ণয় করি (Differentiating Fractional Powers)

$$\frac{d}{dx}(2x^{3/2} - 3x^{4/3} - 5) = 2 \left(\frac{3}{2} x^{\frac{3}{2}-1} \right) - 3 \left(\frac{4}{3} x^{\frac{4}{3}-1} \right) - 0 = 3x^{1/2} - 4x^{1/3}$$

STEP 5 মান বসিয়ে চূড়ান্ত বীজগাণিতিক অন্তরক পাই (Final Form)

$$\frac{dy}{dx} = \frac{5}{2}(3x^{1/2} - 4x^{1/3})(2x^{3/2} - 3x^{4/3} - 5)^{3/2} = \frac{5}{2}(3\sqrt{x} - 4\sqrt[3]{x})(2x^{3/2} - 3x^{4/3} - 5)^{3/2}$$

FINAL ANSWER
$$\frac{dy}{dx} = \frac{5}{2}(3x^{1/2} - 4x^{1/3})(2x^{3/2} - 3x^{4/3} - 5)^{3/2}$$

🔴 BOX 1: FORMULATION & FRACTIONAL EXPONENT RULES

- **Power Rule:** $\frac{d}{dx}(x^p) = px^{p-1}$ for any rational power $p \in \mathbb{Q}$.
- For $x^{3/2}$: $\frac{d}{dx}(x^{3/2}) = \frac{3}{2}x^{1/2} = \frac{3}{2}\sqrt{x}$.
- For $x^{4/3}$: $\frac{d}{dx}(x^{4/3}) = \frac{4}{3}x^{1/3} = \frac{4}{3}\sqrt[3]{x}$.
- Outer power reduction: $\frac{5}{2} - 1 = \frac{3}{2}$.

⚠️ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Arithmetic Mistakes in Exponent Subtraction:** Writing $\frac{4}{3} - 1 = \frac{1}{2}$ instead of $\frac{1}{3}$, or $\frac{5}{2} - 1 = \frac{1}{2}$ instead of $\frac{3}{2}$.
- **Forgetting to Cancel Coefficients:** $2 \times \frac{3}{2} = 3$ and $3 \times \frac{4}{3} = 4$. Missing these cancellations leaves unsimplified fractions.

🌐 BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- **Even Root Constraints:** The terms $x^{3/2}$ and the outer power $(\dots)^{5/2}$ require real square root operations.
- We require $x \geq 0$ and $2x^{3/2} - 3x^{4/3} - 5 \geq 0$.
- Since $2x^{3/2} - 3x^{4/3} - 5 = 0$ has a unique positive real root $x_0 \approx 5.62$, the real domain of y is $[x_0, \infty)$.
- The derivative $\frac{dy}{dx}$ is differentiable on (x_0, ∞) .

💡 BOX 4: LEIBNIZ SUBSTITUTION (ALTERNATIVE VERIFICATION)

- Let $u = 2x^{3/2} - 3x^{4/3} - 5 \implies y = u^{5/2}$.
- $\frac{dy}{du} = \frac{5}{2}u^{3/2}$.
- $\frac{du}{dx} = 3x^{1/2} - 4x^{1/3}$.
- By Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{5}{2}(3x^{1/2} - 4x^{1/3})(2x^{3/2} - 3x^{4/3} - 5)^{3/2}$$

📊 BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 4 / 5 MARKS)

- **1.0 Mark:** Correct application of outer power rule $\frac{5}{2}u^{3/2}$.
- **2.0 Marks:** Precise term-by-term differentiation of fractional exponents $3x^{1/2} - 4x^{1/3}$.
- **1.0 Mark:** Correct combination into final boxed algebraic form.